

Diabetes Prediction — End-to-End ML Analysis

Dataset

The dataset used is the **Pima Indians Diabetes Dataset**, a well-known medical dataset containing **768 patient records** and **9 variables** collected from female patients of Pima Indian heritage. The target variable is Outcome — whether a patient has diabetes (1) or not (0).

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body mass index
DiabetesPedigreeFunction	Diabetes hereditary likelihood score
Age	Age in years
Outcome	1 = Diabetic, 0 = Not diabetic

Step 1 — Missing Value Detection & Imputation

Several columns contained **biologically impossible zero values** — for example, a glucose level of 0 or a BMI of 0 is medically meaningless. These zeros were therefore treated as **hidden missing values (NaN)** rather than true measurements.

The affected columns were: Glucose, BloodPressure, SkinThickness, Insulin, and BMI.

Fix applied: Zeros were replaced with NaN, then imputed using the **group-wise median per Outcome class** (diabetic vs non-diabetic). This approach is more accurate than a global median because the two groups have meaningfully different distributions — for example, diabetic patients tend to have higher glucose levels.

Insulin had the highest proportion of missing values (~49%), while BMI had the least (~1.4%).

Step 2 — Outlier Detection & Treatment

Outliers were detected using the **IQR (Interquartile Range) method**:

- Lower fence = $Q1 - 1.5 \times IQR$
- Upper fence = $Q3 + 1.5 \times IQR$

Any value falling outside these fences was flagged as an outlier.

Fix applied: Winsorisation (capping) - outliers were clipped to the fence values rather than removed. This preserves all 768 rows while eliminating the distorting effect of extreme values on model training.

Insulin and DiabetesPedigreeFunction had the most extreme outliers across all columns.

Step 3 — Exploratory Data Analysis (EDA)

EDA was conducted to understand the structure and patterns in the cleaned data:

- **Class balance:** The dataset is mildly imbalanced - 65.1% non-diabetic vs 34.9% diabetic. This was addressed by using `class_weight='balanced'` in the Random Forest model.
- **Glucose** showed the strongest visual separation between the two classes, making it the most informative single feature.
- **BMI and Age** also showed clear distributional differences between diabetic and non-diabetic patients.
- **Correlation analysis** revealed positive relationships between Glucose ↔ Insulin, Age ↔ Pregnancies, and BMI ↔ SkinThickness — all clinically sensible.
- A **Glucose vs BMI scatter plot** confirmed that diabetic patients tend to cluster toward higher values of both features simultaneously.

Step 4 — Feature Engineering

Six new features were created to give the models richer signals beyond the raw measurements:

New Feature	How It Was Created	Clinical Rationale
Glucose_Risk	Cut into 3 tiers: Normal / Pre-diabetic / Diabetic range	Mirrors clinical glucose classification
BMI_Cat	Cut into 4 categories: Underweight / Normal / Overweight / Obese	BMI categories are clinically meaningful
Age_Group	Cut into 4 bands: Young / Middle / Senior / Elderly	Diabetes risk increases with age
Glucose_BMI	$\text{Glucose} \times \text{BMI}$ (interaction term)	Captures combined metabolic risk
Insulin_Glucose	$\text{Insulin} \div (\text{Glucose} + 1)$	Proxy for insulin resistance
Preg_Age	$\text{Pregnancies} \div (\text{Age} + 1)$	Pregnancy burden relative to age

These engineered features — especially `Glucose_BMI` — turned out to be among the top predictors in the final models.

Step 5 — Model Training & Evaluation

Three classification models were trained on an **80/20 train-test split** with **stratified sampling** to preserve class proportions:

Model	Accuracy	F1 Score	ROC-AUC	5-Fold CV AUC
Logistic Regression	75.97%	0.661	0.844	0.878 ± 0.019
Random Forest	87.01%	0.811	0.937	0.939 ± 0.012
Gradient Boosting	87.01%	0.811	0.960	0.944 ± 0.012

Evaluation metrics used:

- **Accuracy** — overall correct predictions
- **F1 Score** — balance between precision and recall (important for imbalanced classes)
- **ROC-AUC** — ability to discriminate between diabetic and non-diabetic patients across all thresholds
- **5-Fold Cross-Validation AUC** — model stability and generalisability

Winner: Gradient Boosting achieved the highest ROC-AUC of **0.960**, meaning it correctly ranks a diabetic patient above a non-diabetic one 96% of the time. Its low CV standard deviation (± 0.012) also confirms it generalises well and is not overfitting.

Top predictive features (from Random Forest importance): Glucose, Glucose_BMI, BMI, Age, and Insulin - all consistent with established clinical risk factors for Type 2 diabetes.

Summary

The pipeline successfully transformed a raw, messy medical dataset into a robust, production-ready diabetes classifier through systematic data cleaning, exploratory analysis, domain-informed feature engineering, and rigorous model comparison. The final Gradient Boosting model achieves near-clinical-grade discrimination with a **ROC-AUC of 0.96**.